

Operating System

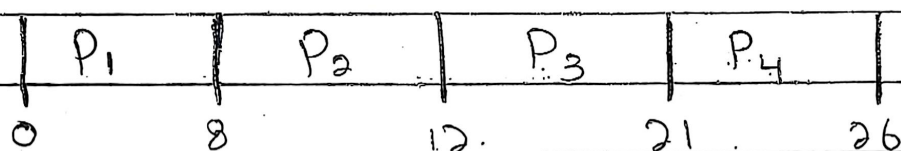
Assignment 1

Saim Zafar

01-135241-045

① First come First serve =

• Gantt Chart =



• Table = (Completion time) (CT).

Process.	(AT) Arrival Time	Burst Time	Completion Time
P ₁	0	8	8
P ₂	1	4	12
P ₃	2	9	21
P ₄	3	5	26

• Turn Around Time Table = (TAT)

Process	CT	AT	TAT
P ₁	8	0	8
P ₂	12	1	11
P ₃	21	2	19
P ₄	26	3	23

• Waiting Time Table =

Process	TAT	BT	WT
P ₁	8	8	0
P ₂	11	4	7
P ₃	19	9	10
P ₄	23	5	18

• Avg WT = $\frac{0+7+10+18}{4} = 8.75$

• Avg TAT = $\frac{8+11+19+23}{4} = 15.25$

② Round Robin = (Time Slice = 3)

Process	AT	BT
P ₁	0	8 5 2 0
P ₂	1	4 1 0
P ₃	2	9 6 8 0
P ₄	3	8 2 0

• Time 0-3 =

P₁ runs → remaining 5

• Time 3-6 =

P₂ runs → remaining 1

• Time 6-9 =

P₃ runs → remaining 6

• Time 9-12 =

P₄ runs → remaining 2

• Time 12-15 =

P₁ runs → remaining 2

• Time 15 - 16 =

P_2 runs $\rightarrow$ Finished.

• Time 16 - 19 =

P_3 runs $\rightarrow$ remaining 3

• Time 19 - 21 =

P_4 runs $\rightarrow$ Finished

• Time 21 - 23 =

P_1 runs $\rightarrow$ Finished.

• Time 23 - 26 =

P_3 runs $\rightarrow$ Finished.

• Gantt Chart =

P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_4	P_1	P_3	
0	3	6	9	12	15	16	19	21	23	26

• CT Table =

Process	CT
P_1	23
P_2	16
P_3	26
P_4	21

• TAT Table

Process	CT	AT	TAT
P_1	23	0	23
P_2	16	1	15
P_3	26	2	24
P_4	21	3	18

• Waiting Time (WT) Table =

Process	TAT	BT	WT
P_1	23	8	15
P_2	15	4	11
P_3	24	9	15
P_4	18	5	13

- Avg $WT = \frac{15+11+15+13}{4} = 13.5$

- Avg $TAT = \frac{23+15+24+18}{4} = 20.$

③ Shortest Remaining Time First = (SRT F)

Process	AT	BT
P ₁	0	8/10
P ₂	1	4/0
P ₃	2	9/0
P ₄	3	5/0

- Time 0-1 = P₁ run → remaining 7

- Time 1 = (P₂ = 4) < (P₁ = 7)
P₂ runs

- Time 1-5 = (P₂ = 4) < (P₃ = 9) and (P₄ = 5)
P₂ runs and Finishes.

[P₃ arrives at T=2.
P₄ arrives at T=3.]

- Time 5 = (P₄ = 5) < (P₁ = 7) and (P₃ = 9)
P₄ runs

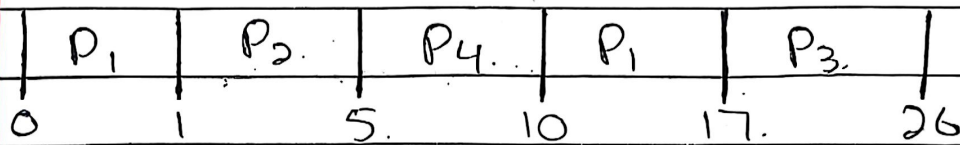
- Time 5-10 = P₄ runs and Finishes

- Time 10 = (P₁ = 7) < (P₃ = 9)
P₁ runs

- Time 10-17 = P₁ runs and Finishes

- Time 17-26 = P₃ runs and Finishes

• Grantt Chart =



• CT, TAT, WT Table =

Process	CT	AT	TAT	BT	WT
P ₁	17	0	17	8	9
P ₂	5	1	4	4	0
P ₃	26	2	24	9	15
P ₄	10	3	7	5	2

• $\text{Avg WT} = \frac{9 + 0 + 15 + 2}{4} = 6.5$

• $\text{Avg TAT} = \frac{17 + 4 + 24 + 7}{4} = 13$

④ SJF (Non-Preemptive) =

Process	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

- Time 0 = 8 = P₁ arrives and executes completely as it starts once and completes even if smallest process arrives
- At T = 1 → P₂ arrives
- At T = 2 → P₃ arrives
- At T = 3 → P₄ arrives

- Time 8 = ($P_2=4$) < ($P_3=9$) and ($P_4=5$)
 P_2 runs.

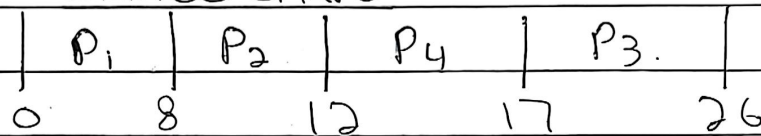
- Time 8-12 = P_2 runs and Finishes.

- Time 12 = ($P_4=5$) < ($P_3=9$)
 P_4 runs.

- Time 12-17 = P_4 runs and Finishes

- Time 17-26 = P_3 runs and Finishes

- Gantt Chart =



- CT, TAT, WT Table =

Process	CT	AT	TAT	BT	WT
P_1	8	0	8	8	0
P_2	12	1	11	4	7
P_3	26	2	24	9	15
P_4	17	3	14	5	9

- Avg WT = $\frac{0+7+15+9}{4} = 7.75$

- Avg TAT = $\frac{8+11+24+14}{4} = 14.25$

⑤ Multilevel Queue Scheduling = (Time Slice=3)

Process	AT	BT	Priority
P ₁	0	8	2
P ₂	1	4	1
P ₃	2	9	3
P ₄	3	7	2

- Queue 1 = P₂ (Highest)
- Queue 2 = P₁, P₄ (Middle)
- Queue 3 = P₃ (lowest)

• - Queue 1 = P₂ • - Queue 2 = P₁, P₄ • - Queue 3 = P₃

• Time 0-1 = P₁ arrives → runs → remaining = 7

• Time 1 = P₂ arrives.

(P₂ priority = 1) < (P₁ priority = 2)
Now P₂ runs

• Time 1-2 = P₂ runs → remaining = 3

• Time 2 = P₃ arrives.

(P₃ priority = 3) > (P₂ priority = 1)

• Time 2-3 = P₂ runs → remaining = 2

• Time 3 = P₄ arrives

(P₄ priority = 2) > (P₂ priority = 1)
P₂ runs

• Time 3-5 = P₂ runs and Finishes

• Time 5 = Queue 1 Finished

Now Queue 2 runs.

• Time 5-8 = P₁ runs → remaining = 4

• Time 8-11 = P₄ runs → remaining = 2

• Time 11-14 = P₁ runs → remaining = 1

- Time 14-16 = P₄ runs and Finishes.

- Time 16-17 = P₁ runs and Finishes

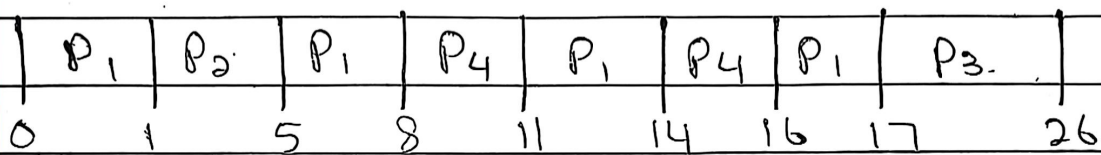
Queue 2 Finished

- Time 17-26 = P₃ runs and Finishes

(FCFS) used for Queue 3.

Queue 3 Finished

- Gantt Chart =



- CT, TAT, WT Table =

Process	CT	AT	TAT	BT	WT
P ₁	17	0	17	8	9
P ₂	5	1	4	4	0
P ₃	26	2	24	9	15
P ₄	16	3	13	5	8

- Avg WT = $\frac{9+0+15+8}{4} = 8$

- Avg TAT = $\frac{17+4+24+13}{4} = 14.5$

- Comparative Analysis =

- SRTF has minimum avg waiting time of 0.5.

- Round Robin is the most fair as it provides equal time slice for CPU.

- SRTF and Multilevel Queue may cause starvation if short jobs keep arriving in SRTF and if lower

priority queues may never execute.

- SRTF are better at performance for shorter jobs. as shortest ones are executed first.
- Round Robin is better for interactive systems. as it has regular CPU sharing.

→ Comparison (Preemptive vs Non-Preemptive) =

→ Preemptive =

- A process may be interrupted if higher priority or shortest job arrives.
- Provides better response time.
- Has overhead more due to frequent context switching.
- More complex to implement.

→ Non-Preemptive =

- Once process starts execution, it can't be interrupted.
- May cause long waiting time if a long process is running.
- Has less overhead due to fewer context switching.
- Simple and easy to implement.